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DRISHTI Innovation Challenge

PARINAM

Platform for Analysis of Related Impacts in Naval Assembly Modifications

Powered by CATALYST Architecture

Proposed Solution · Annexure 1

Problem Statement-70

Field	Details
Applicant Name	Leejo Purvathan
Company Name	Arcaisys
Challenge	PROBLEM STATEMENT-70
Duration	20 Months
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1. Brief Summary of the Proposed Solution

Naval shipbuilding programmes are among the most complex engineering endeavours, involving thousands of interdependent components spanning structural, electrical, propulsion, weapons, and outfitting domains. The DRISHTI Innovation Challenge has identified a critical pain point that imposes significant cost and schedule penalties on Indian naval acquisition programmes. MDL faces significant cost overruns and schedule delays due to the lack of a systematic approach for analysing the impact of design changes across highly interconnected systems. Current methods rely on manual coordination and fragmented assessments, making them slow, inconsistent, and difficult to scale.

Problem Statement-70 addresses Design Change Impact Analysis: when a change is made, such as a cable tray relocation or bulkhead modification, the impact cascades across multiple ship systems, causing 15–35% cost overruns and significant schedule delays. Current assessment workflows are manual, fragmented, and non-scalable.

OVERVIEW OF THE PROPOSED SYSTEM

PARINAM (Platform for Analysis of Related Impacts in Naval Assembly Modifications) is proposed as a unified, on-premise engineering decision-support system designed to systematically analyse the cascading impact of design changes in complex naval shipbuilding programmes. The solution directly addresses the limitations of current manual, fragmented workflows by introducing a structured, repeatable, and auditable approach to impact assessment.

System-Level Modelling of Naval Assets

At its core, PARINAM models the entire ship as an interconnected system of components, where each element (structural, electrical, mechanical, and outfitting) is digitally represented and linked through defined dependency relationships. When a design change is introduced, the system captures the change in a structured format and uses this interconnected model to trace how the modification propagates across related systems. This ensures that all downstream effects, including secondary and tertiary impacts, are systematically identified rather than being dependent on manual expertise.

Structured Change Processing and Impact Traceability

The solution operates through a coordinated pipeline of input processing, dependency analysis, validation, and decision generation. Design changes can be submitted either as natural language inputs or structured data, which are standardised into a consistent internal representation. The system then evaluates the change against a comprehensive dependency model to identify all affected components and systems, providing full traceability of how and why each impact occurs.

Compliance Validation and Engineering Rule Enforcement

In parallel, PARINAM enforces engineering and regulatory compliance by validating all impacted components against predefined rules derived from classification standards and engineering constraints. Any violations are clearly identified along with their severity and required corrective actions, ensuring that compliance risks are not overlooked during the decision-making process.

Cost, Schedule, and Risk Assessment

To support programme-level decision-making, the system further evaluates the financial, schedule, and operational implications of the change. It generates cost estimates based on material, labour, rework, and indirect factors, and assesses potential schedule delays by analysing affected activities within the project.



timeline. Additionally, it provides a consolidated risk score derived from impact severity, constraint violations, and system-wide propagation effects, enabling prioritisation and informed approvals.

Unified Decision Intelligence Interface

All outputs are presented through a unified dashboard interface that offers a clear and structured view of impacted systems, compliance status, cost implications, schedule effects, and overall risk. The system also maintains a complete audit trail of inputs, analyses, and decisions, ensuring transparency, traceability, and accountability in line with defence-grade requirements.

KEY INNOVATION

A key innovation of PARINAM lies in its combination of dependency-driven analysis with rule-based validation and Advanced Deep Learning Intelligence. This ensures that outputs are both systematic and defensible, while reducing reliance on manual expertise. The system also supports standardised input processing and maintains full traceability of all impact pathways and decisions.

Overall Impact and Value Proposition

Overall, PARINAM transforms design change impact analysis from a manual, experience-driven activity into a deterministic, data-driven process. It enables faster, more reliable, and fully traceable decision-making, significantly reducing manual effort, minimising cost and schedule overruns, improved accuracy in cost and schedule estimation, and providing actionable intelligence to engineers and programme managers.

2. Key Technologies Used

#	Technology	Role
1	Neo4j Graph Database	Core knowledge representation , directed dependency graph with weighted edges; BFS-based impact propagation across 10,000+ component nodes
2	BERT (Fine-tuned NLP)	Additional natural language parsing module for structured input generation; system supports structured inputs independently
3	XGBoost	Assisted cost estimation refinement layer (activated when sufficient historical data is available)
4	PERT + Critical Path Method	Schedule impact estimation using a project DAG with $TE=(O+4M+P)/6$; identifies critical path nodes affected by change
5	Random Forest Classifier	Complimentary statistical risk scoring support; primary risk assessment derived from deterministic factors
6	Drools Rule Engine	Deterministic constraint validation against IEC 61084, structural load standards, and naval engineering rules



3. Deliverables

S.No	Deliverable Name	Type	Brief Description
D-01	Digital Twin Core	Data + Engine	Structured component data model integrating BOM, engineering attributes, and rule bindings (extensible to CAD integration where available)
D-02	Dependency Knowledge Graph	Database (Neo4j)	Directed weighted graph encoding all inter-system dependencies with documented relationship taxonomy.
D-03	Impact Propagation Engine	Software Module	BFS/DFS traversal with exponential depth decay; produces ranked impact lists with propagation path explanations.
D-04	Decision Output Engine	Software Module	Consolidated engineer-facing report: compliance status, impact analysis, cost estimates, and risk scores.
D-05	System Integration Package	APIs + Docs	REST API layer, PLM adapter stubs, CAD system webhooks, and full OpenAPI 3.0 specification.
D-06	Security & Deployment Package	Infrastructure	Docker-Compose/Kubernetes manifests for air-gapped on-premise deployment.
D-07	Validation & Acceptance Report	Document	End-to-end system validation against real engineering scenarios; accuracy metrics; SME sign-off certificate.
D-08	NLP Change Request Parser	Software Module	Fine-tuned BERT NLP model and REST API accepting free-text/structured design change requests, outputting validated JSON change objects.
D-09	Constraint Validation Engine	Module + Rule Library	Rule engine with domain-specific engineering constraints: cable tray fill capacity (IEC 61084), structural loads, thermal clearances, pressure ratings.
D-10	Two-Layer Cost Estimation Engine	Module + ML Model	Layer 1: Deterministic formula. Layer 2: XGBoost model trained on historical records with 80% and 95% confidence intervals. Historical estimate-vs-actual delta as a 'variance forecast indicator'
D-11	Schedule Impact Module (PERT/CPM)	Software Module	PERT/CPM engine on project DAG; computes delay in working days/weeks, float



S.No	Deliverable Name	Type	Brief Description
			consumption, and schedule recovery recommendations.
D-12	Risk Scoring Engine	Module + Trained Model	Hybrid risk scoring module combining deterministic indicators with Random Forest classifier producing composite risk score (0–1) with SHAP-based explainability listing top 3 risk drivers.
D-13	Decision Intelligence Dashboard	Web App (On-Premise)	Secured role-based dashboard: impacted systems map, constraint violations, cost estimates, Gantt impact view, risk gauge, PDF export.

4. Proposed Timeline (20 Months)

Phase	Month(s)	Key Activities	Phase Output
Phase 0 , Foundation	1–3	Stakeholder interviews; data audit; engineering constraint catalogue; Neo4j schema design; NLP training corpus collection; IT infrastructure assessment.	Signed-off data schema and constraint catalogue
Phase 1 , Core Engines	4–8	Knowledge graph construction; BFS propagation engine; BERT NLP fine-tuning; Drools rule engine; Digital Twin data model; deterministic cost estimation layer.	All core PARINAM engines operational in dev
Phase 2 , ML & Optimization	9–11	XGBoost cost model training; Random Forest risk classifier; SHAP explainability layer; end-to-end pipeline integration; model retraining pipeline design.	Integrated ML pipeline with full end-to-end outputs
Phase 3 , Dashboard & Integration	12–13	Decision Intelligence Dashboard UI; RBAC implementation; PDF report engine; REST API gateway; PLM adapter stubs; security hardening.	Secured on-premise dashboard deployed
Phase 4 , Pilot & Validation	14–16	Pilot scenarios will be constructed from MDL historical design change records (Projects 15B, 17A, 75) provided under data-sharing agreement. System outputs will be validated against actual recorded cost and schedule outcomes from MDL programme archives; UAT with programme managers.	Validated accuracy vs. historical actuals; UAT sign-off



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Phase	Month(s)	Key Activities	Phase Output
Phase 5 , Handover & Docs	17–18	Final model retraining; system documentation; 2-day user training workshop; Field Operations Manual; source code handover.	Full system handover with acceptance certificate
Phase 6 , Production Enhancements	19–20	Engineering Change Workflow (ECO): formal 6-stage lifecycle (Proposal → Impact Analysis → SME Review → Approval → Execution → Closure) with PKI-based digital signatures at each approval gate and immutable audit log. Zero-trust security: mTLS between all microservices; RBAC extended with attribute-based controls. Collaborative engineering: multi-user concurrent sessions with conflict detection and version locking.	Production-grade enterprise system deployed

5. Project Milestones

M	Milestone Name	Month	Completion Criteria	Deliverables	Phase Total (₹)
M-01	Foundation Complete	3	Data schema signed off; constraint catalogue approved; dev environment operational. Minimum 200 historical change records confirmed available or transfer learning plan agreed with MDL	Schema Docs, Rule Catalog	₹24,00,000
M-02	Core Engines Ready	8	All core engines (graph, NLP, rule engine, Digital Twin) pass unit and integration tests.	D-01 to D-04, D-08	₹60,00,000
M-03	ML Pipeline Integrated	11	ML models trained and validated; full end-to-end pipeline from input to output functional.	D-09 to D-12	₹48,00,000
M-04	Dashboard Deployed	13	Secured on-premise dashboard with RBAC; PDF report generation functional; security audit passed.	D-05 to D-06, D-13	₹39,00,000
M-05	Pilot Validation Passed	16	All pilot scenarios executed; cost estimate $\pm 15\%$, risk classification $\geq 85\%$ accuracy; validated concurrence with engineering assessments across pilot scenarios, with acceptable variance in cost and impact estimation ,UAT sign-off.	Pilot Report, UAT Sign-off	₹60,00,000
M-06	System Handover	18	Full source code delivered; documentation complete; 2-day	D-07, FOM, Training Report	₹30,00,000



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M	Milestone Name	Month	Completion Criteria	Deliverables	Phase Total (₹)
			training conducted; acceptance report signed.		
M-07	Production System Live	20	All Phase 6 enhancements deployed; zero-trust security validated; ECO workflow live. Independent security penetration test passed; system demonstrated on live MDL environment with programme manager sign-off	Production Deployment Report	₹39,00,000
TOTAL PROJECT COST					₹3,00,00,000
*This estimate is based on current project requirements and is subject to change upon any scope modifications, unforeseen technical complexities, or third-party fee adjustments with the requirement that funds for each milestone must be released within 2-3 weeks of Milestone completion.					

6. Abbreviations & Technical Glossary

The following table defines all abbreviations and technical terms used throughout this document.

Abbreviation / Term	Full Form / Explanation
ABS	American Bureau of Shipping , one of the international ship classification societies whose rules PARINAM validates against.
AI	Artificial Intelligence , the broader field enabling PARINAM's intelligent impact analysis and risk scoring.
API	Application Programming Interface , a defined contract by which software modules communicate; PARINAM exposes all capabilities via REST APIs.
BERT	Bidirectional Encoder Representations from Transformers , a transformer-based NLP model fine-tuned by PARINAM to parse free-text design change requests.
BFS	Breadth-First Search , the graph traversal algorithm used by the Impact Propagation Engine to discover all components affected by a design change.
BOM	Bill of Materials , the structured list of all components, sub-assemblies, and raw materials constituting a ship system.
BV	Bureau Veritas , French international classification society; one of the rule sources in the PARINAM compliance engine.
CAD	Computer-Aided Design , software used to create and modify digital 3D models of ship components and assemblies.
PARINAM	Platform for Analysis of Related Impacts in Naval Assembly Modifications , Arcaisys's proposed system for Problem Statement-70.



Abbreviation / Term	Full Form / Explanation
CATALYST	Change Analysis Through Linked Systems and Intelligent Tracking, the underlying engine architecture framework upon which PARINAM is implemented.
CPM	Critical Path Method , a schedule analysis technique identifying the longest sequence of dependent tasks, used by PARINAM to compute minimum project duration after a change.
DAG	Directed Acyclic Graph , a graph structure with directed edges and no cycles; used to represent the project schedule in PARINAM's PERT/CPM module.
DFS	Depth-First Search , an alternative graph traversal strategy used alongside BFS in the impact propagation engine for deep dependency chains.
DNV	Det Norske Veritas , Norwegian classification society; one of the rule sources in the PARINAM compliance engine.
DRISHTI	Defence Research and Innovation for Strategic and Technology Harnessing Initiative , the GoI defence innovation challenge under which this proposal is submitted.
ECO	Engineering Change Order , a formal, multi-stage workflow for approving, implementing, and documenting design changes in a naval programme.
ERP	Enterprise Resource Planning , business management software (e.g., SAP, Oracle) integrated with PARINAM for cost rate table ingestion and BOM synchronisation.
FOM	Field Operations Manual , the operational documentation delivered to the end-user organisation enabling independent operation and maintenance of PARINAM.
GPU	Graphics Processing Unit , hardware accelerator used to train ML models (XGBoost, Random Forest) significantly faster than CPU-only compute.
IEC 61084	International Electrotechnical Commission standard 61084 governing cable tray systems; PARINAM validates cable fill ratios against this standard.
IRS	Indian Register of Shipping , India's national ship classification society; the primary rule source for PARINAM's compliance engine.
JSON	JavaScript Object Notation , a lightweight data interchange format; PARINAM's NLP parser outputs structured JSON change objects.
LDAP / AD	Lightweight Directory Access Protocol / Active Directory , enterprise directory services used by PARINAM's RBAC module for user authentication.
LR	Lloyd's Register , British classification society; one of the rule sources in the PARINAM compliance engine.
MAWP	Maximum Allowable Working Pressure , the maximum internal pressure a pipe or pressure vessel is designed to withstand; validated by PARINAM's constraint engine.



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Abbreviation / Term	Full Form / Explanation
MinIO	An open-source, S3-compatible object storage system used within PARINAM's on-premise deployment for storing CAD files, model artifacts, and audit records.
ML	Machine Learning , a sub-field of AI; PARINAM uses ML for cost prediction (XGBoost), risk scoring (Random Forest), and constraint violation pattern detection.
mTLS	Mutual Transport Layer Security , a security protocol requiring both client and server to authenticate each other; used in PARINAM's Phase 6 zero-trust architecture.
NER	Named Entity Recognition , an NLP sub-task used by PARINAM's BERT parser to extract component names, locations, change types, and magnitudes from free-text inputs.
Neo4j	A leading graph database management system; used as the storage and query engine for PARINAM's Dependency Knowledge Graph.
NLP	Natural Language Processing , a branch of AI enabling computers to understand and process human language; used by PARINAM to parse design change requests.
OpenAPI 3.0	A widely adopted specification standard for describing REST APIs; PARINAM's system integration package includes a full OpenAPI 3.0 specification.
PERT	Programme Evaluation and Review Technique , a probabilistic scheduling method using optimistic, most-likely, and pessimistic duration estimates: $TE = (O + 4M + P) / 6$.
PKI	Public Key Infrastructure , a cryptographic framework used for digital signatures in PARINAM's Engineering Change Order approval workflow.
PLM	Product Lifecycle Management , software systems (e.g., Siemens Teamcenter, PTC Windchill) managing engineering data across a product's lifecycle; PARINAM provides PLM adapter stubs.
RBAC	Role-Based Access Control , a security model restricting system access based on user roles (Viewer, Analyst, Approver, Admin) as implemented in PARINAM's dashboard.
REST	Representational State Transfer , an architectural style for designing networked APIs using standard HTTP methods; all PARINAM interfaces are REST-based.
SAP / Oracle	Major ERP software vendors; PARINAM's integration layer includes connectors for SAP PM and Oracle for BOM and cost data synchronisation.
SHAP	SHapley Additive exPlanations , a game-theory-based method for explaining ML model predictions; PARINAM uses SHAP to provide human-readable risk drivers and cost factors.
SME	Subject Matter Expert , domain specialists (naval engineers, programme managers) engaged during pilot validation and constraint catalogue development.



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Abbreviation / Term	Full Form / Explanation
UAT	User Acceptance Testing , a validation phase where end-users test the system against real-world scenarios to confirm it meets operational requirements.
UUID	Universally Unique Identifier , a 128-bit identifier used by PARINAM to uniquely tag every component in the Digital Twin and knowledge graph.
XGBoost	eXtreme Gradient Boosting , an ensemble ML algorithm used in PARINAM's two-layer cost estimation engine, trained on historical naval design change records.
XML / CSV	eXtensible Markup Language / Comma-Separated Values , structured data formats used for BOM ingestion, schedule export, and PLM data interchange in PARINAM.